AI and Quantitative Methods

LLMs and Quantitative Research in Social and Political Sciences

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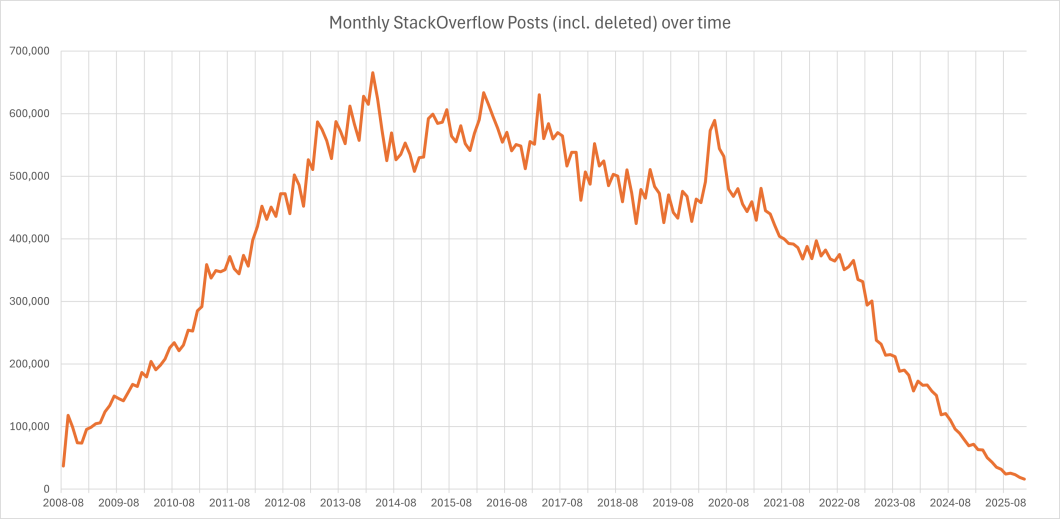
Session Materials

All code, examples, and this document are available online.

- Document (.qmd) with all code
- Extended examples for reference
- Links to packages, papers, and tools

`github.com/patrickeshea/llm-quant-polisci`


Monthly StackOverflow Posts (incl. deleted) over time
700,000
600,000
500,000
400,000
300,000
200,000
100,000
0
2008-08
2009-08
2010-08
2011-08
2012-08
2013-08
2014-08
2015-08
2016-08
2017-08
2018-08
2019-08
2020-08
2021-08
2022-08
2023-08
2024-08
2025-08

What This Session Covers

In scope today

- LLM-assisted coding in R
- Text-as-data applications
- Open vs. closed model trade-offs
- Validation and reproducibility

Deliberately excluded

- Writing assistance
- Literature review tools
- Teaching applications
- Ethics debates

Separate conversations worth having — just not today.

AI-Assisted Coding: Three Levels of Autonomy

Autonomy

High

Claude Code — agentic coding from the command line

Multi-step tasks: download data, merge, run analysis, iterate on errors.

Requires Anthropic subscription or API. **Live demo today.**

GitHub CoPilot in RStudio — inline autocomplete

Watches what you type, suggests completions. Free via university GitHub Pro.

Low

LLM Chat — paste an error or a question

Claude, ChatGPT, Gemini. Immediate help, but manual copy-paste workflow.

Text-as-Data with LLMs

LLMs process meaning and context from text. This is a major step forward for political science applications.

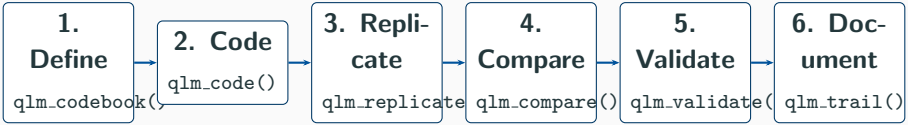
What LLMs can do

- **Classify**: immigration policy, sentiment, topic
- **Scale**: left–right positioning from text
- **Code open-ended responses**: thematic categorisation
- **Beyond text**: images, audio

Key references

- Gilardi et al. 2023 (*PNAS*)
- Halterman & Keith 2026 (*Pol. Analysis*)
- Benoit et al. 2026 (*AJPS*)
- Törnberg 2025 (*SSCR*)

The `quallmer` Workflow



- Codebook-driven annotation
- GPT-4o-mini as primary coder
- Llama 3.2:1b (local, open) as replication
- Inter-coder reliability (Krippendorff's α, Cohen's κ)
- Validation against human gold standard
- Full audit trail for replication archives

Maerz & Benoit 2026. **Live demo today.**

Open vs. Closed Models

	Closed (GPT, Claude, Gemini)	**Open** (Llama, Mixtral, Qwen)
Data sensitivity	Data sent to provider servers	Runs locally; data stays on your machine
Performance	Generally stronger on complex tasks	Competitive on narrow tasks; improving fast
Reproducibility	Provider can update/retire models	You control the exact model version
Cost	Pay per token via API	Free to run; requires compute (GPUs)
Setup	Minimal — sign up for API key	Higher — local infrastructure needed

Practical default: closed models via API for most researchers. Switch to open when data is sensitive, reproducibility is critical, or transparency is required.

If this was helpful..

These tools are developing fast. Three things we can do as a School:

1. **Share what works.** Build a common reference point for which tools, packages, and workflows are useful for our research
2. **Develop shared infrastructure.** API access, local model hosting, and data protection guidance are easier to solve collectively.
3. **Keep talking.** Best practices are still being worked out in our discipline. Regular conversations — across methods, career stages, and subfields — will help us stay ahead of the curve.

Next Event: 13 May

Automating Workflows, Maintaining Integrity

- 13 May 2026, 12:00–17:00
- 130 Hetherington Building

AI ethics & integrity, workflow automation, advanced methods for social science. Places limited — sign up to express interest. Organised by Dan Xu (SPS PGR).

Supported by the MacFie SPS Fund.

Sign up here

Thank you!